

AI meets Geometry

XXII ESCOLA DE GEOMETRIA DIFERENCIAL
FOUR-LECTURE MINICOURSE

Learn how AI can approximate geometric structures, solve variational PDEs, and search for new mathematical examples.



1 AI Overview

Map supervised, unsupervised, and reinforcement learning onto geometric approximation, pattern-finding, and search.



2 Learning Geometric Structures

Train neural networks to learn metrics and differential forms while enforcing symmetry, positivity, and geometric identities.



3 Neural PDEs & Geometric Flows

Encode geometric PDEs in neural losses and flow an immersed torus toward the Willmore minimiser.



4 Discovery & Verification

Search line arrangements and other geometric spaces with reinforcement learning, then verify discoveries exactly.

>_ **HANDS-ON PYTHON THROUGHOUT** Starter code is provided; no prior Python experience is assumed.